

Alpha5 Gating

Goal

Under Output Stationary Mode (OS Mode), when a zero input operand is delivered to a PE, the corresponding registers will not sample the zero input, entailing no dynamic power on MAC, which could be energy efficient when inferring a model with high sparsity.

How to Run

Compile *.v files within the *filelist* and run simulation with *core_tb_output_stationary.v*. Add a random *mac_tile* instance signals to wave, to observe the wave format.

Register *a_q* should not toggle when *x_zero_in* is set, and register *b_q0* and *b_q1* should not change when *w_zero_in* is set.

